

Calculus for Biology and Medicine, 3rd Edition, Claudia Neuhauser

10.5 #1, 3, 6, 9, 15, 16, 17, 18, 25, 36, 42

1) $f(x, y) = x^2 + y^2, x(t) = 3t, y(t) = e^t, t = \ln 2$

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = 6x + 2ye^t$$

$$\frac{\partial f}{\partial x} = 2x \quad \frac{\partial f}{\partial y} = 2y \quad x(\ln 2) = 3(\ln 2) = 3\ln 2$$

$$\frac{dx}{dt} = 3 \quad \frac{dy}{dt} = e^t \quad y(\ln 2) = e^{(\ln 2)} = 2$$

$$\frac{dw}{dt}(\ln 2) = 6(3\ln 2) + 2(2)e^{(\ln 2)} = \boxed{18\ln 2 + 8}$$

3) $f(x, y) = \sqrt{x^2 + y^2}, x(t) = t, y(t) = \sin t, t = \frac{\pi}{3}$

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = \frac{x + y \cos t}{\sqrt{x^2 + y^2}}$$

$$\frac{\partial f}{\partial x} = \frac{x}{\sqrt{x^2 + y^2}} \quad \frac{\partial f}{\partial y} = \frac{y}{\sqrt{x^2 + y^2}} \quad x\left(\frac{\pi}{3}\right) = \left(\frac{\pi}{3}\right) = \frac{\pi}{3}$$

$$\frac{dx}{dt} = 1 \quad \frac{dy}{dt} = \cos t \quad y\left(\frac{\pi}{3}\right) = \sin\left(\frac{\pi}{3}\right) = \frac{\sqrt{3}}{2}$$

$$\frac{dw}{dt}\left(\frac{\pi}{3}\right) = \frac{\left(\frac{\pi}{3}\right) + \left(\frac{\sqrt{3}}{2}\right) \cos\left(\frac{\pi}{3}\right)}{\sqrt{\left(\frac{\pi}{3}\right)^2 + \left(\frac{\sqrt{3}}{2}\right)^2}} = \boxed{\frac{4\pi + 3\sqrt{3}}{2\sqrt{4\pi^2 + 27}}}$$

6) $f(x, y) = xe^y, x(t) = e^t, y(t) = t^2, t = 0$

$$\frac{dw}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} = e^y(e^t + 2xt)$$

$$\frac{\partial f}{\partial x} = e^y \quad \frac{\partial f}{\partial y} = xe^y \quad x(0) = e^{(0)} = 1$$

$$\frac{dx}{dt} = e^t \quad \frac{dy}{dt} = 2t \quad y(0) = (0)^2 = 0$$

$$\frac{dw}{dt}(0) = e^{(0)}(e^{(0)} + 2(1)(0)) = \boxed{1}$$

15) $r(F(t), N(t))$

$$\frac{dr}{dt} = \frac{\partial r}{\partial F} \frac{dF}{dt} + \frac{\partial r}{\partial N} \frac{dN}{dt}$$

$$< 0$$

Growth rate
decreases over time.

$\frac{\partial r}{\partial F} > 0$ Growth rate increases
with availability of food.

$\frac{\partial r}{\partial N} < 0$ Growth rate decreases
with number of competitors.

$\frac{dF}{dt} < 0$ Availability of food
decreases over time.

$\frac{dN}{dt} > 0$ Number of competitors
increases over time.

$$16) S(T(x), P(x))$$

S - abundance of plant species

T - temperature

P - precipitation

x - distance traveled

$$\frac{\partial S}{\partial T} > 0 \quad \text{Abundance increases with temperature.}$$

$$\frac{\partial S}{\partial P} > 0 \quad \text{Abundance increases with precipitation.}$$

$$\frac{dT}{dx} > 0 \quad \text{Temperature increases along path.}$$

$$\frac{dP}{dx} > 0 \quad \text{Precipitation increases along path.}$$

$$\frac{dS}{dx} = \frac{\partial S}{\partial T} \frac{dT}{dx} + \frac{\partial S}{\partial P} \frac{dP}{dx} > 0$$

You would expect to encounter this species more often during your journey.

$$17) f(x, y) = x^3 y^2$$

$$\nabla f(x, y) = \begin{bmatrix} 3x^2 y^2 \\ 2x^3 y \end{bmatrix}$$

$$18) f(x, y) = \frac{xy}{x^2 + y^2}$$

$$\nabla f(x, y) = \begin{bmatrix} \frac{(x^2 + y^2)(y) + (xy)(2x)}{(x^2 + y^2)^2} \\ \frac{(x^2 + y^2)(x) + (xy)(2y)}{(x^2 + y^2)^2} \end{bmatrix}$$

$$25) f(x, y) = \sqrt{2x^2 + y^2}, (1, 2), \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$\nabla f(x, y) = \begin{bmatrix} \frac{2x}{\sqrt{2x^2 + y^2}} \\ \frac{y}{\sqrt{2x^2 + y^2}} \end{bmatrix}$$

$$u = \frac{1}{\sqrt{(1)^2 + (1)^2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{aligned} D_u f(1, 2) &= \frac{2(1)}{\sqrt{2(1)^2 + (2)^2}} \cdot \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \\ &= \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{6}} \left(\begin{bmatrix} 2 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right) \\ &= \frac{1}{\sqrt{12}} ((2)(1) + (2)(1)) \\ &= \frac{4}{\sqrt{12}} = \frac{2}{\sqrt{3}} \end{aligned}$$

$$36) f(x, y) = e^x \cos y, (0, \frac{\pi}{2})$$

$$\nabla f(x, y) = \begin{bmatrix} e^x \cos y \\ -e^x \sin y \end{bmatrix}$$

$$\nabla f(0, \frac{\pi}{2}) = \begin{bmatrix} e^0 \cos(\frac{\pi}{2}) \\ -e^0 \sin(\frac{\pi}{2}) \end{bmatrix} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$42) f(x, y) = xy, (2, 3)$$

$$\nabla f(x, y) = \begin{bmatrix} y \\ x \end{bmatrix} \quad \nabla f(2, 3) = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$

$$\frac{1}{\sqrt{(3)^2 + (2)^2}} \begin{bmatrix} 3 \\ 2 \end{bmatrix} = \frac{1}{\sqrt{13}} \begin{bmatrix} 3 \\ 2 \end{bmatrix}$$